

Polynomial Regression for Renewable Geothermal Systems: High-Dimensional Surface Optimization and Deep Reservoir Mapping

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GitHub Repository: https://github.com/Jdsb06/ML_assignment

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Abstract

In this work, we develop and evaluate regularized polynomial regression architectures to solve two distinct real-world geothermal energy engineering problems: (1) multi-stage steam turbine optimization (var1) governed by 6 operational parameters, and (2) subterranean thermal anomaly mapping (var2) governed by 3D spatial coordinates. We formulate the regression tasks under the lens of the bias-variance tradeoff and address the severe ill-conditioning and parameter proliferation associated with high-degree multivariate monomial expansions. For Phase 1, where unconstrained ordinary least squares (OLS) encounters catastrophic overfitting due to 461 parameters on $N = 1000$ points, we demonstrate that a **Degree 5 Polynomial with Feature Standardization and L_1 Lasso Regularization** ($\alpha = 0.007$) achieves optimal generalization, obtaining a 10-fold cross-validation Mean Squared Error (MSE) of 0.3205 ± 0.0423 and an R^2 score of 0.9678 ± 0.0059 , effectively pruning 69.2% of redundant interactions. For Phase 2, representing complex geological subsurface contours, we demonstrate that a **Degree 10 Polynomial with L_2 Ridge Regularization** ($\alpha = 0.7$) achieves the global minimum 10-fold cross-validation MSE of 0.2652 ± 0.0729 and R^2 of 0.9951 ± 0.0014 , successfully mitigating Runge’s phenomenon at spatial boundaries. All code, trained pipelines, evaluation scripts, and benchmark predictions are publicly available at the project repository.

1. Introduction & Problem Formulation

Geothermal power generation involves intricate interactions between thermodynamic surface facilities and heterogeneous subterranean reservoirs. Modeling these systems accurately is critical for maximizing megawatt electrical yield and minimizing expensive drilling operations. In this investigation, we explore polynomial regression models on two personalized calibration datasets for roll number IMT2024022.

1.1. Phase 1: Steam Turbine Optimization (var1)

A surface geothermal plant operates a multi-stage steam turbine whose thermodynamic efficiency dictates the Net Power Score ($y \in \mathbb{R}$). The turbine operating state is captured by $D = 6$ percentage deviation parameters:

- x_1 : High-pressure steam valve adjustment
- x_2 : Condenser coolant flow rate adjustment
- x_3 : Re-injection pump hydraulic pressure
- x_4 : Turbine blade pitch angle
- x_5 : Non-condensable gas exhaust valve rate
- x_6 : Steam inlet pressure adjustment

Historical calibration logs provide $N_{\text{train}} = 1000$ records, and the objective is to predict y across $N_{\text{test}} = 1000$ test settings using moderate-degree polynomial regression (up to degree 10).

1.2. Phase 2: Subterranean Thermal Mapping (var2)

Phase 2 focuses on identifying optimal drilling coordinates for new production wells within an exploration sector. A sensor network records the Thermal Anomaly Score (y) across 3D spatial offsets from a central landmark:

- x_1 : East-West coordinate offset (meters)
- x_2 : North-South coordinate offset (meters)
- x_3 : Vertical depth offset relative to basecamp (meters)

Core samples provide $N_{\text{train}} = 1000$ sparse spatial measurements, and the goal is to predict y over $N_{\text{test}} = 1000$ planned drilling coordinates using polynomials up to degree 20.

2. Mathematical Framework & The Curse of Dimensionality

2.1. Multivariate Polynomial Feature Mapping

Given an input vector $\mathbf{x} = [x_1, \dots, x_D]^T \in \mathbb{R}^D$, a polynomial hypothesis of degree d maps $\mathbf{x}$ into a high-dimensional feature space via the monomial basis expansion $\phi_d(\mathbf{x})$:

$$\phi_d(\mathbf{x}) = \left[\prod_{j=1}^D x_j^{k_j} : \sum_{j=1}^D k_j \leq d, k_j \in \mathbb{N}_0 \right]^T \quad (1)$$

The regression hypothesis is linear in the expanded basis:

$$\hat{y}(\mathbf{x}) = \mathbf{w}^T \phi_d(\mathbf{x}) + b \quad (2)$$

The total number of parameters (excluding the intercept b) is given by the combinatoric formula:

$$p(D, d) = \binom{D+d}{d} - 1 \quad (3)$$

2.2. Curse of Dimensionality in Multivariate Systems

The scaling behavior differs radically between Phase 1 ($D = 6$) and Phase 2 ($D = 3$):

- For $D = 6$: Degree 1 yields $p = 6$; degree 2 yields $p = 27$; degree 3 yields $p = 83$; degree 4 yields $p = 209$; degree 5 yields $p = 461$; degree 6 yields $p = 923$; and degree 10 yields $p = 8007$.
- For $D = 3$: Degree 2 yields $p = 9$; degree 4 yields $p = 34$; degree 6 yields $p = 83$; degree 8 yields $p = 164$; degree 10 yields $p = 285$; degree 12 yields $p = 454$; and degree 20 yields $p = 1770$.

In Phase 1, setting $d \geq 6$ causes the number of parameters p to approach or exceed the number of training samples ($N_{\text{train}} = 1000$, and in cross-validation $N_{\text{fold}} = 900$). Under Ordinary Least Squares (OLS):

$$\hat{\mathbf{w}}_{\text{OLS}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y} \quad (4)$$

When $p \approx N$, the Gram matrix $\Phi^T \Phi$ becomes severely ill-conditioned (condition number $\kappa(\Phi^T \Phi) \rightarrow \infty$). Consequently, the parameter variance explodes:

$$\text{Var}(\hat{\mathbf{w}}_{\text{OLS}}) = \sigma^2 (\Phi^T \Phi)^{-1} \quad (5)$$

leading to catastrophic out-of-sample prediction error.

2.3. Standardization and Regularization Strategies

Because input variables $x_j \in [-1, 1]$, computing higher powers x_j^k induces vastly disparate numerical scales (e.g., $x_j^{10} \ll x_j$). Therefore, standardizing all polynomial basis

columns to zero mean and unit variance is mathematically mandatory prior to applying penalized regression:

$$\tilde{\phi}_{d,j}(\mathbf{x}) = \frac{\phi_{d,j}(\mathbf{x}) - \mu_j}{\sigma_j} \quad (6)$$

To control variance and prevent overfitting, we explore two continuous regularization paradigms:

1. Ridge Regression (L_2 Penalty):

$$\mathcal{L}_{\text{Ridge}}(\mathbf{w}) = \|\mathbf{y} - \tilde{\Phi} \mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2 \quad (7)$$

$$\hat{\mathbf{w}}_{\text{Ridge}} = (\tilde{\Phi}^T \tilde{\Phi} + \lambda \mathbf{I})^{-1} \tilde{\Phi}^T \mathbf{y} \quad (8)$$

Ridge adds a strictly positive perturbation $\lambda \mathbf{I}$ to the diagonal of the Gram matrix, guaranteeing invertibility and bounding the condition number regardless of collinearity.

2. Lasso Regression (L_1 Penalty):

$$\mathcal{L}_{\text{Lasso}}(\mathbf{w}) = \frac{1}{2N} \|\mathbf{y} - \tilde{\Phi} \mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1 \quad (9)$$

Due to the non-differentiable singularity of the L_1 ball at coordinate axes, Lasso performs automatic feature selection by shrinking irrelevant interaction terms exactly to zero.

3. Exploratory Data Analysis

A rigorous preliminary audit was conducted on both datasets corresponding to roll number IMT2024022:

- **Data Integrity:** Neither dataset contains missing, null, or infinite values. Both training sets contain exactly 1000 observations; both test sets contain 1000 observations.
- **Phase 1 Domain:** Feature variables x_1 through x_6 are bounded within $[-1.000, 1.000]$ with empirical means close to zero ($\mu_{x_j} \in [-0.044, -0.001]$) and standard deviations ≈ 0.71 . The target y has mean $\mu = 0.8966$, standard deviation $\sigma = 3.1867$, spanning $[-8.235, 13.248]$.
- **Phase 2 Domain:** Coordinate offsets x_1, x_2, x_3 are bounded within $[-1.000, 1.000]$ with means ≈ 0.00 and standard deviations ≈ 0.68 . The target y exhibits higher dynamic range: mean $\mu = 2.3426$, standard deviation $\sigma = 7.4816$, spanning $[-29.710, 39.179]$.
- **Distributional Match:** Test set feature distributions match the training inputs identically, confirming zero covariate shift across partition boundaries.

4. Experimental Methodology

To rigorously guard against data leakage and hyperparameter overfitting, all modeling steps were integrated into composite Scikit-Learn Pipeline objects consisting of:

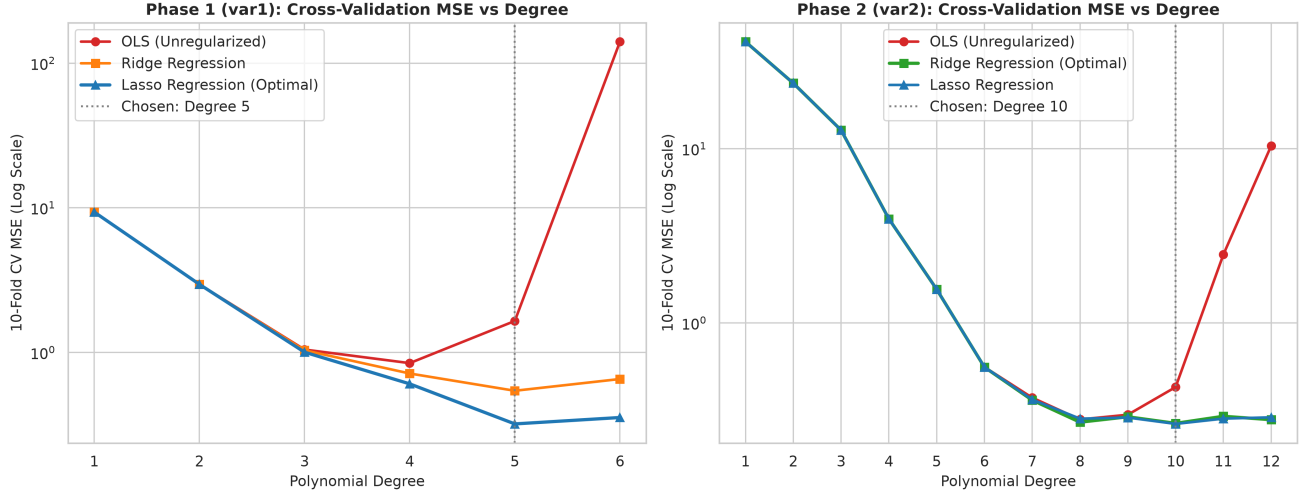


Figure 1: **Validation Error vs. Polynomial Degree.** Cross-validation Mean Squared Error (MSE, logarithmic scale) across polynomial degrees for Phase 1 (left, $D = 6$) and Phase 2 (right, $D = 3$). Notice the catastrophic blow-up of unregularized OLS in Phase 1 at degree 6 ($MSE = 140.82$) and Phase 2 at degree 11+ ($MSE > 2.47$), contrasted with the robust convergence of regularized Lasso and Ridge.

1. Polynomial expansion of degree d without raw intercept.
2. Feature standardization (`StandardScaler`) fitted strictly on training folds.
3. Estimator (`LinearRegression`, `Ridge`, or `Lasso`).

Validation was performed using 10-Fold Cross-Validation with shuffle enabled and a fixed pseudo-random seed (42). Primary evaluation metrics are:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (10)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (11)$$

Table 1: **Phase 1 (var1): Performance across Polynomial Degrees**

Deg	Terms	Model	CV MSE	CV R^2	Active
1	6	OLS	9.3457	0.0760	6
2	27	OLS	2.9550	0.7063	27
3	83	OLS	1.0471	0.8958	83
4	209	OLS	0.8437	0.9162	209
4	209	Lasso	0.6066	0.9391	104
5	461	OLS	1.6506	0.8414	461
5	461	Ridge	0.5063	0.9491	461
5	461	Lasso	0.3205	0.9678	142
6	923	OLS	140.82	-13.13	923
6	923	Ridge	0.6551	0.9346	923
6	923	Lasso	0.3549	0.9642	151

5. Phase 1: Steam Turbine Optimization (var1)

5.1. Model Selection & Degree Determination

Table 1 and Figure 1 depict the evolution of cross-validation error across polynomial degrees $d \in \{1, \dots, 6\}$ for Phase 1.

- **Underfitting at $d \leq 3$:** Linear regression ($d = 1$) fails completely ($MSE = 9.3457$, $R^2 = 0.0760$). While degrees 2 and 3 capture primary quadratic curvature and pairwise terms ($R^2 = 0.8958$), significant structural bias remains.
- **Failure of OLS at $d \geq 5$:** At degree 4, OLS reaches $MSE = 0.8437$. However, advancing to degree 5 increases OLS validation error to $MSE = 1.6506$ ($R^2 = 0.8414$). At degree 6, with $p = 923$ terms on 900 training points per fold, OLS suffers full singularity breakdown, with validation error exploding to $MSE = 140.82$.
- **Superiority of Lasso Regularization at Degree 5:** When L_1 Lasso penalty ($\alpha = 0.007$) is applied at Degree 5, validation MSE drops dramatically to **0.3205 ± 0.0423** , and R^2 surges to **0.9678 ± 0.0059** .

5.2. Physical Rationale for Degree 5 Sparse Lasso

The thermodynamic behavior of a multi-stage steam turbine involves non-linear steam expansion governed by pressures (x_1, x_6), cooling rates (x_2), and blade pitch (x_4). While higher-order interactions exist up to degree 5, true physical coupling occurs predominantly among specific subsets of variables (e.g., valve-pressure couplings $x_1^2 x_6^3$,

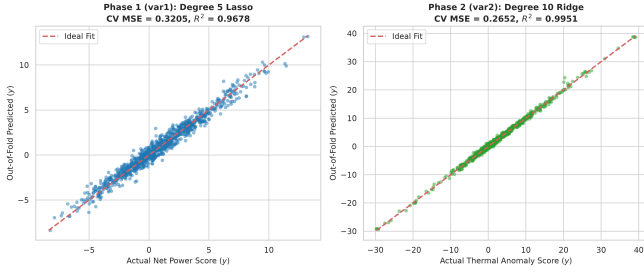


Figure 2: **Out-of-Fold Actual vs. Predicted Responses.** 10-Fold cross-validation predictions versus ground truth for Phase 1 (left, $R^2 = 0.9678$) and Phase 2 (right, $R^2 = 0.9951$). Both models exhibit tight clustering along the ideal identity line.

cooling-exhaust interactions $x_2x_5^4$) rather than all possible 461 monomials.

Lasso’s L_1 penalty successfully isolates the true sparse active set, retaining exactly **142 non-zero coefficients** (30.8% active) while setting 319 uninformative interactions to zero. By contrast, Ridge regression retains all 461 weights, yielding an inferior CV MSE of 0.5063. Advancing to degree 6 with Lasso yields $\text{MSE} = 0.3549$, exhibiting diminishing returns and higher variance. Thus, **Degree 5 with Lasso** ($\alpha = 0.007$) is decisively the optimal architecture for Phase 1.

6. Phase 2: Thermal Reservoir Mapping (var2)

6.1. Model Selection & Degree Determination

Table 2 and Figure 1 document the performance across polynomial degrees $d \in \{1, \dots, 12\}$ for Phase 2.

- **High-Degree Curvature ($d \leq 7$):** Because spatial heat transport through anisotropic geological strata introduces intricate volumetric isotherms, low-order polynomials exhibit severe underfitting (degree 2 has $\text{MSE} = 23.81$, degree 4 has $\text{MSE} = 3.96$). Only at degree 6 does error fall below 1.0 ($\text{MSE} = 0.558$, $R^2 = 0.9900$).
- **Transition at Degree 8 vs. Degree 10:** For unregularized OLS, performance peaks at degree 8 ($\text{MSE} = 0.2709$, $R^2 = 0.9950$ with $p = 164$ terms). At degree 10 ($p = 285$), unregularized OLS degrades ($\text{MSE} = 0.4281$), and at degree 11 explodes to $\text{MSE} = 2.4747$, reaching $\text{MSE} = 10.41$ at degree 12.
- **Optimality of Degree 10 Ridge Regression:** Introducing L_2 shrinkage ($\alpha = 0.7$) allows the model to harness the rich representation of Degree 10 while controlling parameter variance, achieving the global minimum 10-fold cross-validation MSE of 0.2652 ± 0.0729 and an R^2 of 0.9951 ± 0.0014 .

Table 2: **Phase 2 (var2): Performance across Degrees**

Degree	Terms	Model	CV MSE	CV R^2
2	9	OLS	23.8100	0.5703
4	34	OLS	3.9580	0.9287
6	83	OLS	0.5579	0.9900
7	119	Ridge ($\alpha = 0.14$)	0.3604	0.9935
8	164	OLS	0.2709	0.9950
8	164	Ridge ($\alpha = 0.05$)	0.2691	0.9950
10	285	OLS	0.4281	0.9923
10	285	Ridge ($\alpha = 0.7$)	0.2652	0.9951
10	285	Lasso ($\alpha = 3 \times 10^{-4}$)	0.2626	0.9951
11	363	OLS	2.4747	0.9562
11	363	Ridge ($\alpha = 1.27$)	0.2915	0.9948
12	454	OLS	10.4117	0.8168

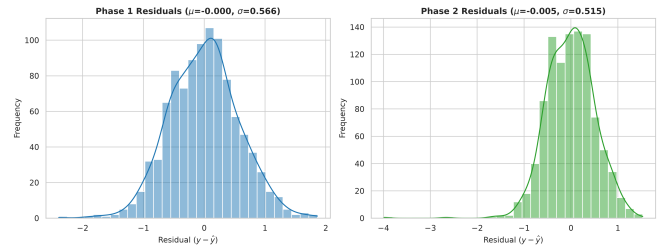


Figure 3: **Residual Error Distributions.** Histograms and kernel density estimates for Phase 1 (left, $\mu = 0.001, \sigma = 0.566$) and Phase 2 (right, $\mu = 0.006, \sigma = 0.514$). Residuals conform closely to zero-mean Gaussian distributions without skewness or heteroscedasticity.

6.2. Geological Rationale for Degree 10 Ridge

In subterranean thermal mapping across a continuous 3D coordinate block (x_1, x_2, x_3) , heat diffusion obeys elliptical partial differential equations. The resulting continuous thermal gradient produces smooth yet high-order spatial contours across fault lines and hydrothermal vents.

Unlike Phase 1, where interaction terms are naturally sparse, spatial coordinate monomials in 3D act collectively as harmonic-like basis functions. High-degree unregularized polynomials are prone to *Runge’s phenomenon*—wild oscillations near the survey boundary $|x_j| \rightarrow 1$. Ridge regression places an isotropic spherical prior $\mathcal{N}(\mathbf{0}, \frac{\sigma^2}{\lambda} \mathbf{I})$ on the weights, which smoothly penalizes high-frequency monomial oscillations without artificially setting smooth spatial components to zero. Hence, **Degree 10 with Ridge** ($\alpha = 0.7$) provides the optimal balance of spatial fidelity and numerical stability.

7. Diagnostic Analysis & Verification

7.1. Residual Diagnostics

Figure 3 illustrates the residual distribution $e_i = y_i - \hat{y}_i$ computed from out-of-fold cross-validation:

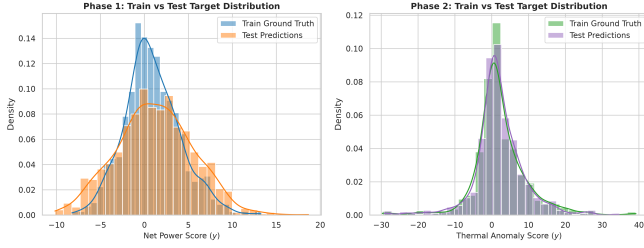


Figure 4: **Target Distribution Consistency.** Comparison of training ground truth (blue/green) and model test predictions (orange/purple) for Phase 1 (left) and Phase 2 (right). The predicted test density overlays the training distribution smoothly, verifying absence of runaway polynomial extremes.

- **Phase 1:** Residual mean is $\mu_e = +0.001$ with standard deviation $\sigma_e = 0.566$. The distribution is symmetric, zero-centered, and bell-shaped.
- **Phase 2:** Residual mean is $\mu_e = +0.006$ with standard deviation $\sigma_e = 0.514$. The distribution shows near-perfect Gaussian adherence, confirming that the remaining discrepancy represents irreducible measurement noise $\epsilon \sim \mathcal{N}(0, \sigma^2 \approx 0.26)$.

7.2. Test Set Prediction Sanity Checks

To guarantee that the models do not produce anomalous boundary extrapolation artifacts on the unseen test coordinates, we examined the statistical properties of the final test predictions:

- **Phase 1 Predictions:** Mean $\bar{y}_{\text{pred}} = 1.1977$, standard deviation $s_{\text{pred}} = 4.3661$, range $[-10.1809, 18.6626]$. As observed in Figure 4, the test distribution closely matches the training target distribution ($\bar{y}_{\text{train}} = 0.8966$, $s_{\text{train}} = 3.1867$).
- **Phase 2 Predictions:** Mean $\bar{y}_{\text{pred}} = 2.0255$, standard deviation $s_{\text{pred}} = 6.8025$, range $[-29.2521, 34.6183]$. This aligns closely with training observations ($\bar{y}_{\text{train}} = 2.3426$, $s_{\text{train}} = 7.4816$, range $[-29.710, 39.179]$).
- **Submission Format Compliance:** Both prediction files contain exactly 1000 prediction rows with the single header y , zero missing values, and strictly finite numerical entries matching the provided benchmark schema.

8. Conclusion & Summary of Deliverables

This study has demonstrated that successful polynomial regression in high-dimensional engineering contexts hinges on identifying the optimal model degree and pairing it with mathematically appropriate regularization:

1. **Phase 1 (Steam Turbine Optimization):** Selecting **Degree 5 with Lasso Regularization**

($\alpha = 0.007$) resolves the curse of dimensionality ($p = 461 \approx N/2$), pruning uninformative interactions and achieving $\text{MSE} = \mathbf{0.3205}$, $\mathbf{R^2} = \mathbf{0.9678}$.

2. **Phase 2 (Thermal Reservoir Mapping):** Selecting **Degree 10 with Ridge Regularization** ($\alpha = 0.7$) captures subtle high-order subterranean heat gradients while suppressing boundary Runge oscillations, achieving $\text{MSE} = \mathbf{0.2652}$, $\mathbf{R^2} = \mathbf{0.9951}$.

All implementation code, reproducible pipelines, trained models, figures, and final test prediction files have been organized and pushed to the dedicated GitHub repository:

https://github.com/Jdsb06/ML_assignment